Workshop Week 4: Extra solutions

Q1

[55,62): half-open interval, 55 included, 62 excluded

(55,62]: half-open interval, 55 excluded, 62 included

(55,62): open interval, 55 and 62 excluded

[55,62]: close interval, 55 and 62 included

Q3 & Q4

Refer to Workshop4Qs34.cpp

Make sure to include <iomanip> header to use setprecision()

Q7 c)

For example: 56+23

Solution 1:

1356: load 56 to R3

1423: load 23 to R4

6534: add two numbers in R3 and R4, then load the sum to R5

45A5: store number in R5 to memory location A5

Solution 2:

Assume 56 is stored at memory location 05; 23 is stored at memory location 06

3305: Load R3 with the value (56) at memory location 05

3406: Load R4 with the value (23) at memory location 06

6534: add two numbers in R3 and R4, then load the sum to R5

45A5: store number in R5 to memory location A5

Q7 d)

1006: load R0 with 6

1506: load 6 to R5

D513: jump to memory address 13 if R5=R0=6

Q8

D0BD: jump to address BD if R0=R0, unconditional jump

BD is stored in program counter

Q9

112A: load 2A to R1

202A: load number in RA to R2

332A: load number at memory address 2A to R3

442A: store number in R4 to memory address 2A

552A: add numbers in R2 and RA, then load sum to R5, for floating point addition

662A: add numbers in R2 and RA, then load sum to R6, for integer addition

882A: shift number in R8 to the right for 2A bits

DD2A: jump to memory address 2A if RD=R0